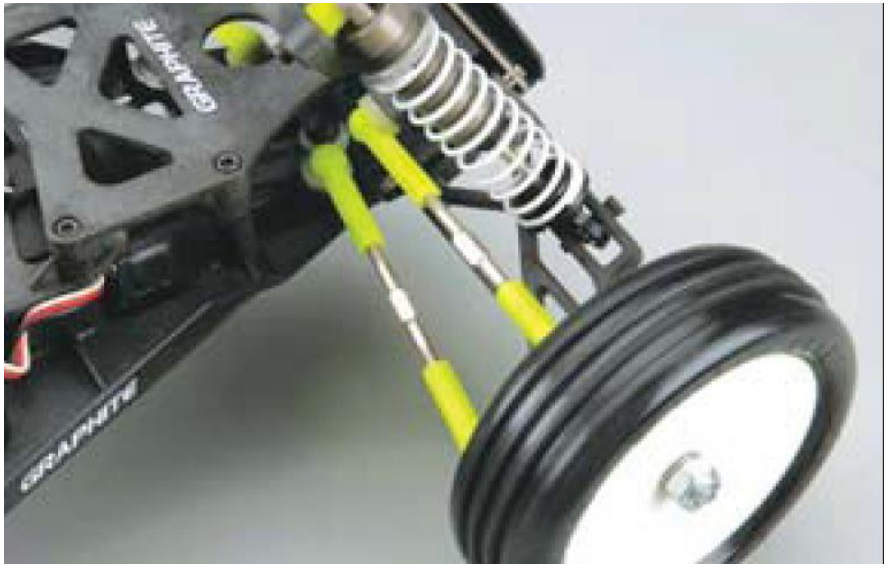
Mechanical vibration





Mechanical vibration

Mechanical vibration is repetitive motion of a particle or system of bodies relative to a stationary reference frame or equilibrium position. Vibrations occur everywhere in machines and structures. The performances of many engineering systems are limited by the vibrational properties of the systems. Vibration can be harmful and in such cases it must be avoided. It can be very useful and desired, or may increase the performance of an engineering system.

Mechanical vibration

Present-day machines and structures often contain high-energy sources which create intense vibration excitation problems, and modern construction methods result in systems with low mass and low inherent damping. Therefore careful design and analysis is necessary to avoid resonance or an undesirable dynamic performance.

Vibration

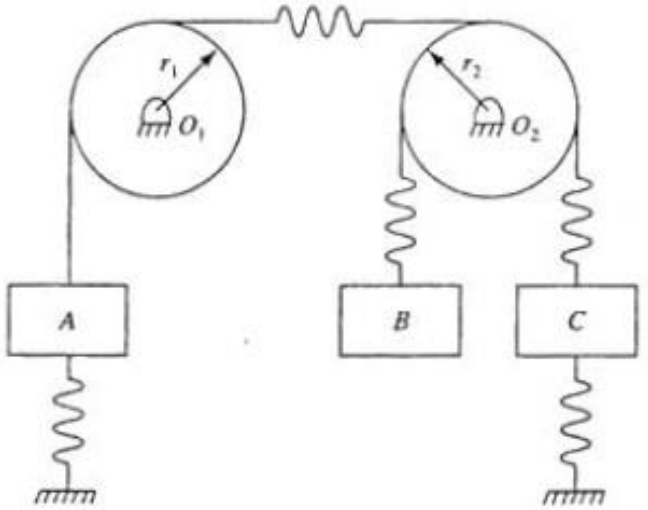
- **Vibration** is a motion repeated at regular intervals. It is expressed in terms of frequency.
- **Cycle** is a single complete repetition of a vibratory motion.
- **Period** is the time required to complete one cycle
- **Frequency** is the number of cycles completed in a given interval of time, usually one second, but occasionally, one minute.
- **Amplitude** is the maximum displacement of a vibrating object from its initial position.
- **Torque** is an action tending to produce rotation of an object.
- **Torsional Vibration** is the twisting and untwisting of a shaft resulting from the periodic application of torque.
- **Damp** is to dissipate energy from a vibrating system.

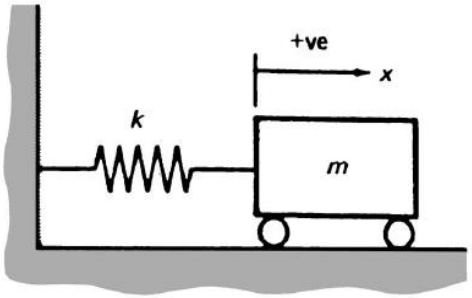
Uses of vibration analysis

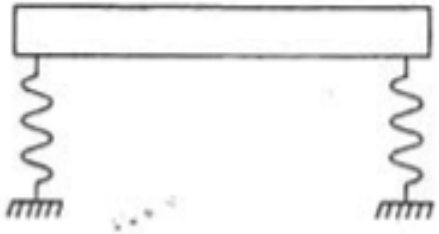
- Preventive maintenance
- Incoming inspection
- Quality control
- Engineering and research
- Plant engineering
- Field service and machinery start-up
- Sales and customer satisfaction
- Noise control
- Balancing
- auscultation

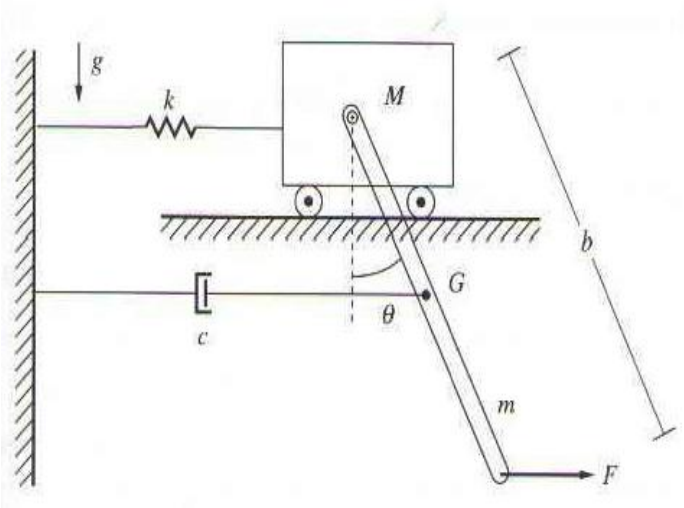
Degree of Freedom

The number of *degrees of freedom* used in the analysis of a mechanical system is the number of kinematically independent coordinates necessary to completely describe the motion of every particle in the system.









n=3N-C

n is degree of freedom

N is number of particles(masses)

C is number of constraints

Dynamic analysis

Dynamic analysis can be carried out most conveniently by adopting the following three-stage approach:

I. MODEL

II. EQUATIONS

III. EXCITATION(RESPONSE)

MECHANICAL SYSTEM COMPONENTS

A mechanical system comprises inertia components, stiffness components, and damping components.

- The inertia components have kinetic energy when the system is in motion. The kinetic energy of a rigid body undergoing planar motion is

$$K.E = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

- A linear stiffness component (a linear spring) has a force displacement relation of the form $F=kx$
- where *F* is applied force and *x* is the component's change in length from its unstreched length.
- The stiffness *k* has dimensions of force per length.





MECHANICAL SYSTEM COMPONENTS

- A *dashpot(damper)* is a mechanical device that adds viscous damping to a mechanical system. A linear viscous damping component has a force-velocity relation of the form; $F=cv$
- where c, is the damping coefficient of dimensions mass per time.







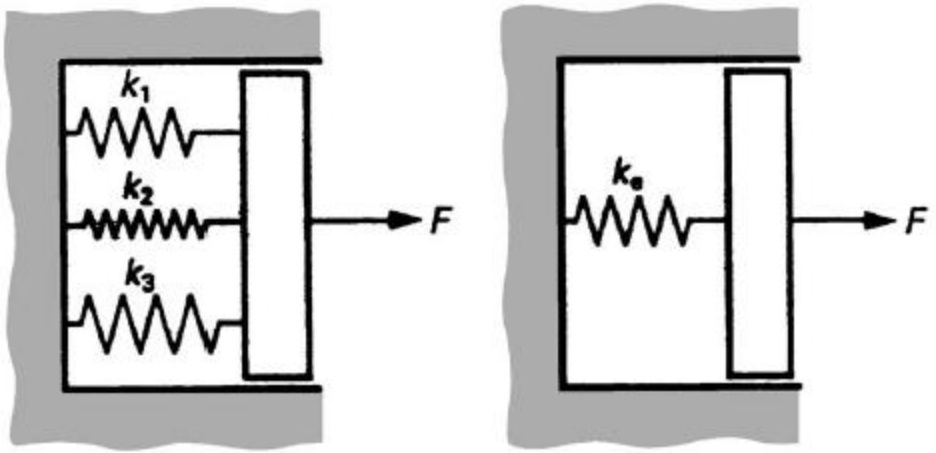
Springs connected in series



$$\delta = \delta_1 + \delta_2 + \delta_3$$

$$\frac{F}{k} = \frac{F}{k_1} + \frac{F}{k_2} + \frac{F}{k_3}; \qquad \frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3}$$

Springs connected in parallel

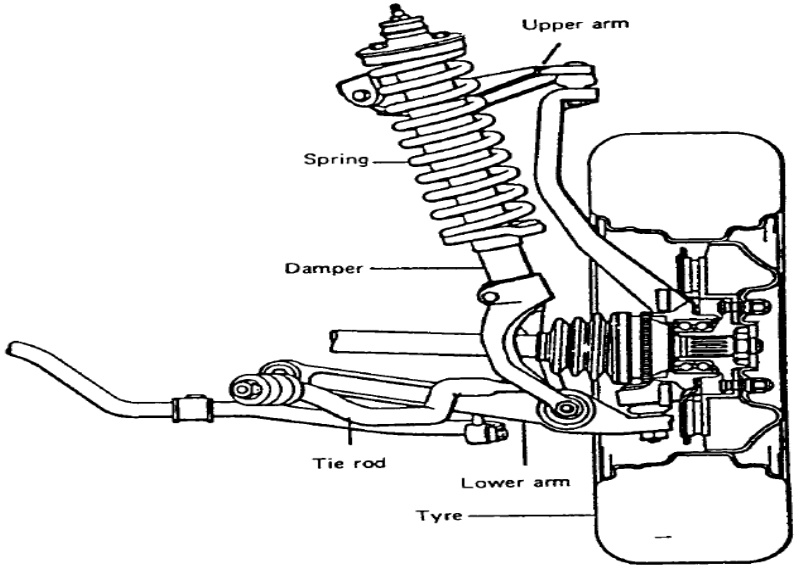


$$F = F_1 + F_2 + F_3$$

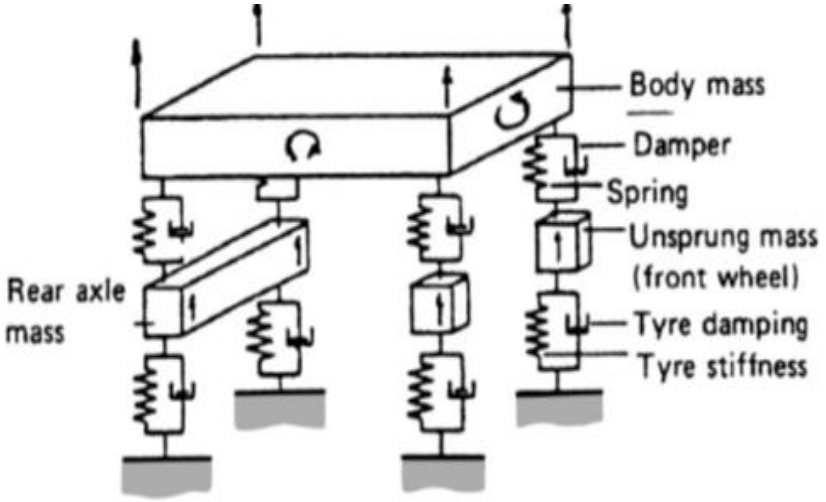
$$k\delta = k_1\delta + k_2\delta + k_3\delta$$

$$k = k_1 + k_2 + k_3$$

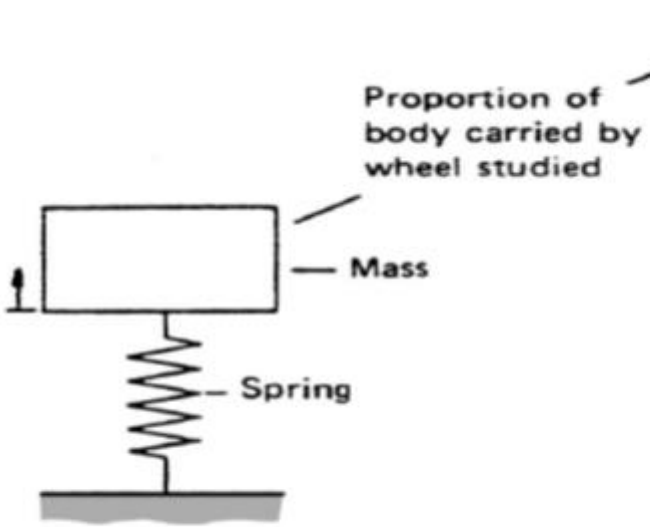
Modeling



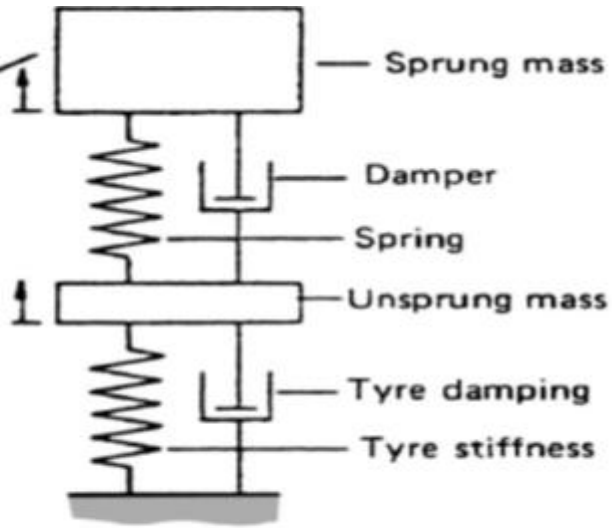
Rover 800 front suspension. (By courtesy of Rover Group.)



Motion in a vertical direction, roll, and pitch can be analysed

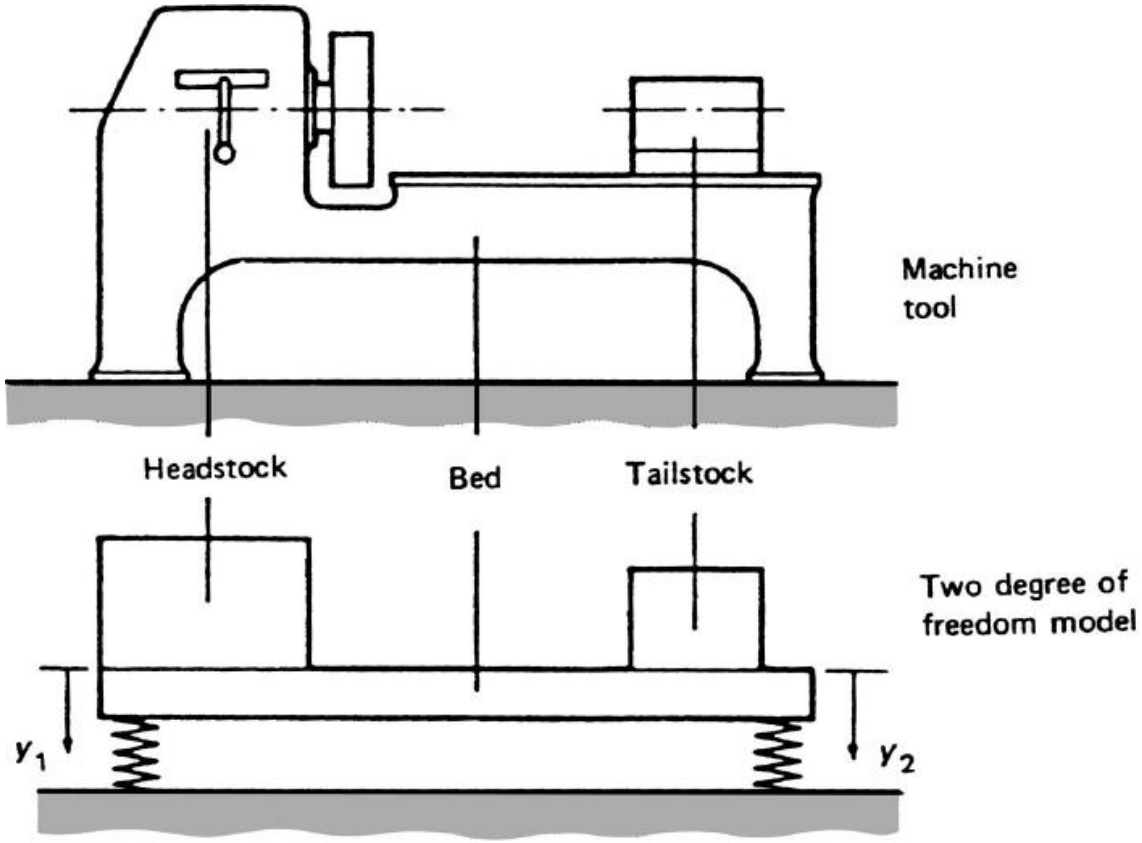


Simplest model – motion in a vertical direction only can be analysed.



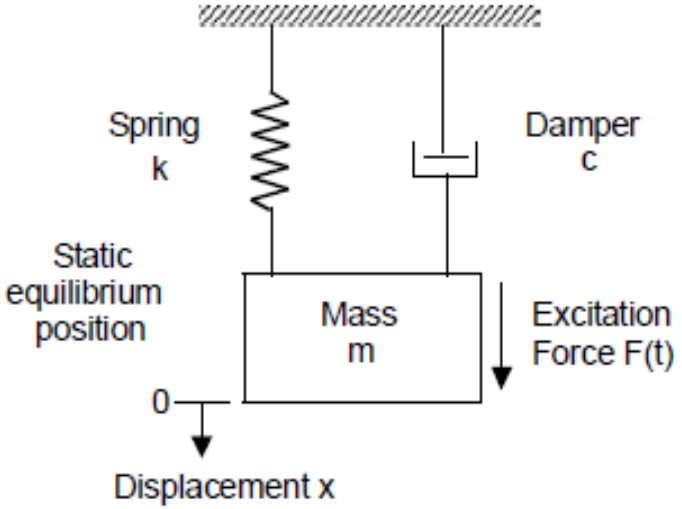
Motion in a vertical direction only can be analysed.

Modeling



Machine tool vibration analysis model.

EQUIVALENT SYSTEMS ANALYSIS



The kinetic energy of a linear system can be written in the form

$$K.E = \frac{1}{2}mv^2$$

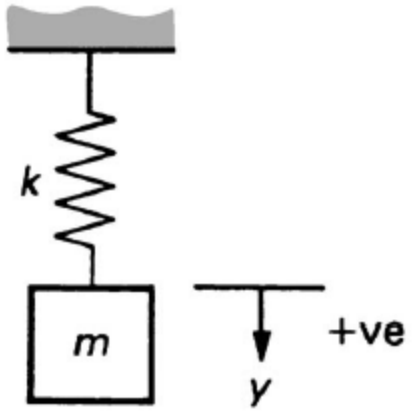
The potential energy of a linear system can be written in the form

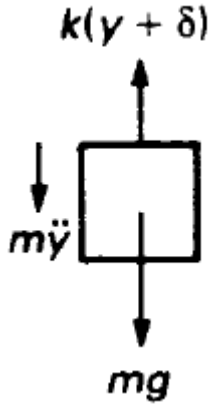
$$P.E = \int kx\, dx = \frac{1}{2}kx^2$$

The work done by the viscous damping force in a linear system between two arbitrary locations x_1, and x_2, can be written as

$$W = \int_{x_1}^{x_2} -cv\, dx$$

FREE UNDAMPED VIBRATION-S.H.M





$$mg = k\delta$$

$$m\ddot{y} = -k(y + \delta) + mg;$$

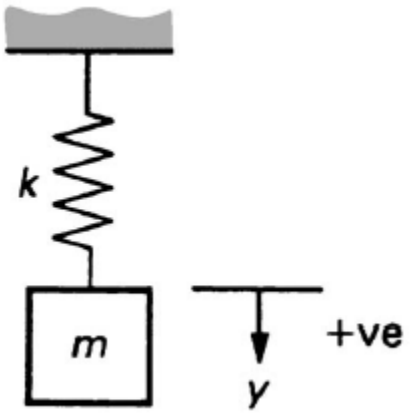
$$\ddot{y} + \left(\frac{k}{m}\right) y = 0;$$

Energy Method

- For undamped free vibration the total energy in the vibrating system is constant throughout the cycle. Therefore the maximum potential energy *V,* is equal to the maximum kinetic energy *T,* although these maxima occur at different times during the cycle of vibration.

$T + V =$ constant, and thus $\frac{d}{dt}(T + V) = 0.$

Example



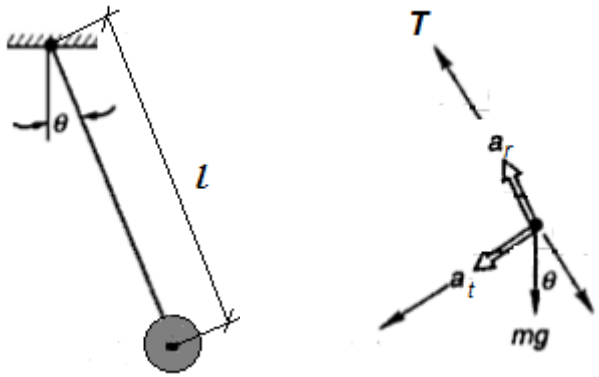
$$T = \frac{1}{2}m\dot{y}^2;\ V = \frac{1}{2}ky^2$$

$$\frac{d}{dt}\left(\frac{1}{2}m\dot{y}^2 + \frac{1}{2}ky^2\right) = 0$$

$$m\ddot{y} + ky = 0$$

The Simple Pendulum

Newtonian Approach



$$\sum M = I\alpha = I\ddot{\theta}$$

$$-ma_t l = I\ddot{\theta};\ a_t = g\sin\theta$$

$$I\ddot{\theta} + mgl\theta = 0$$

$$\ddot{\theta} + \omega^2\theta = 0;\ \ since\ I = ml^2$$

Energy Method

$$P.E = V = mgl(1 - \cos\theta)$$

$$K.E = T = \frac{1}{2}I\dot{\theta}^2$$

$$\frac{d}{dt}(V + T) = 0$$

$$\frac{d}{dt}\left(mgl(1 - \cos\theta) + \frac{1}{2}I\dot{\theta}^2\right) = 0$$

$$I\ddot{\theta} + mgl\theta = 0$$

$$\ddot{\theta} + \omega^2\theta = 0;\ \ since\ I = ml^2$$

$$where\ \ \omega^2 = \frac{g}{l}$$

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{l}{g}}$$

Spring-mass system in which the mass of the spring matters

The kinetic energy of the mass *M* at the mid-position is as before

$$\frac{1}{2}M(\omega_n X)^2$$

The mass of the element δl is $m\frac{\delta l}{L}$, so its kinetic energy at the mid position is $\frac{1}{2}\left(m\frac{\delta l}{L}\right)(\omega_n \frac{l}{L}X)^2$, and of the whole spring is therefore

$$\frac{1}{2}m{\omega_n}^2X^2\frac{1}{L^3}\int_0^L l^2dl = \frac{1}{2}\frac{m}{3}{\omega_n}^2X^2$$

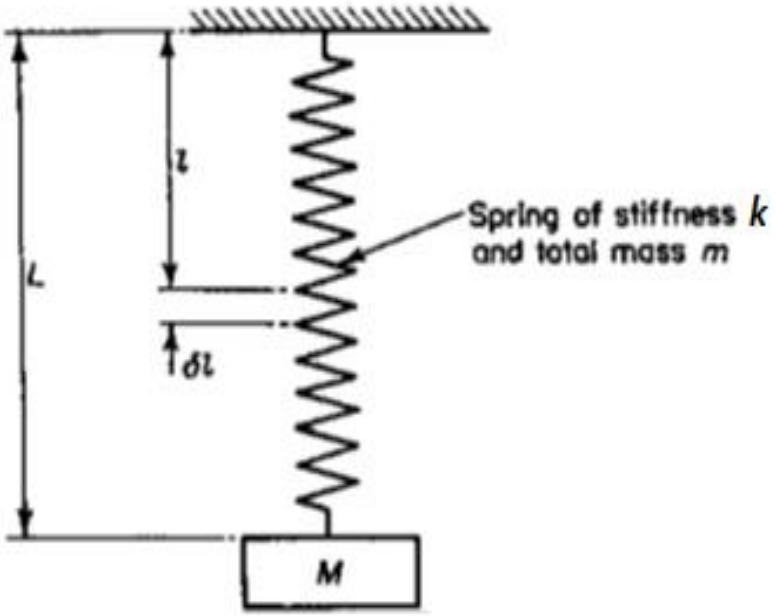
So the total kinetic energy -in the system is

$$\frac{1}{2}\left(M+\frac{m}{3}\right){\omega_n}^2X^2$$

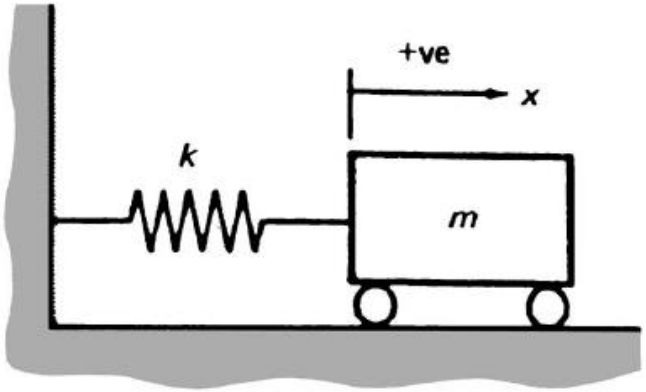
and equating this to the strain energy we get

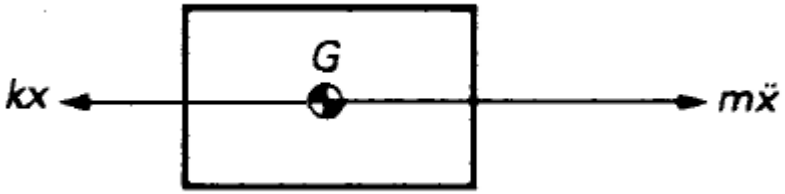
$$\omega_n = \sqrt{\frac{k}{M+\frac{1}{3}m}}\ rad/s$$

$$f_n = \frac{1}{2\pi}\sqrt{\frac{k}{M+\frac{1}{3}m}}\ Hz$$



FREE UNDAMPED VIBRATION





$$m\ddot{x} + kx = 0; \quad \ddot{x} + \left(\frac{k}{m}\right)x = 0;$$

$$\ddot{x} + {\omega_n}^2 x = 0; {\omega_n}^2 = \frac{k}{m}; f = \frac{\omega_n}{2\pi};$$

$$T = 1/f$$

$$x = A sin\omega_n t + B cos\omega_n t$$

Assumed initial conditions

$$x(0) = x_o; \ \dot{x}(0) = v_o$$

$$@\ t = 0; x = x_o$$

$$x_o = B$$

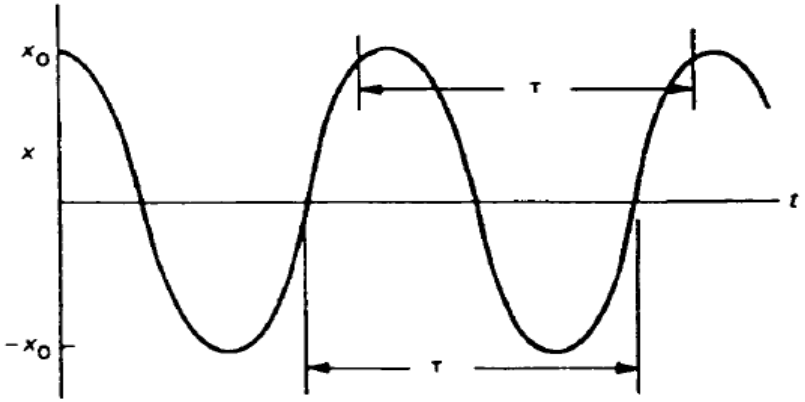
$$@\ t = 0; \dot{x} = v_o$$

$$\dot{x} = A\omega cos\omega_n t - B\omega sin\omega_n t$$

$$v_o = A\omega_n$$

$$A = v_o/\omega_n$$

$$x = (v_o/\omega_n) sin\omega_n t + x_o cos\omega_n t$$



Example 1

Consider a small spring about 30 mm long, welded to a stationary table (ground) so that it is fixed at the point of contact, with a 12-mm bolt welded to the other end which is free to move. The mass of this system is about 49.2 x 10^{-3} kg. The spring stiffness is measured to be k = 857.8 N/m. Calculate the natural frequency and period. Also determine the maximum amplitude of the response if the spring is initially deflected 10 mm.

Solution

The natural frequency is

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{857.8\ N/m}{49.2\ x\ 10^{-3}\ kg}} = 132\ \text{rad/s}$$

In hertz, this becomes

$$f_n = \frac{\omega_n}{2\pi} = 21\ \text{Hz}$$

The period is

$$T = \frac{2\pi}{\omega_n} = \frac{1}{f_n} = 0.0476\ \text{s}$$

Solution

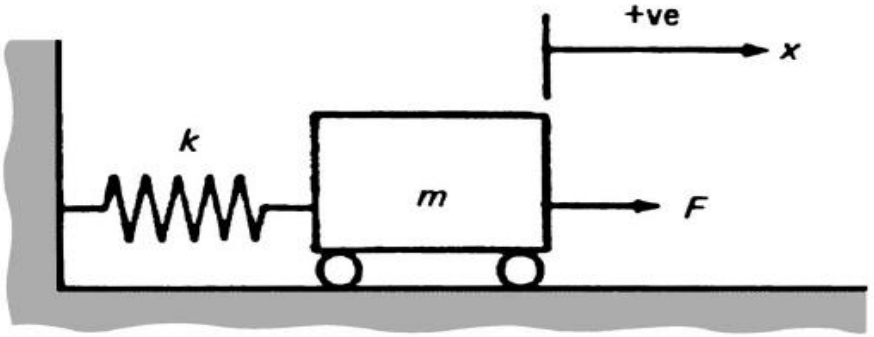
The maximum value of the displacement response,, corresponds to the value of the constant *A*. Assuming that no initial velocity is given to the spring (v_o = 0),

$$x(t)_{max} = A = \frac{\sqrt{\omega_n^2 x_o^2 + \upsilon_o^2}}{\omega_n} = x_o = 10 \text{ mm}$$

The maximum value of the velocity response is $\omega A \; or \; \omega x_o$ = 1320 mm/s and the acceleration response has maximum value $\omega_n^2 A = \omega_n^2 x_o$ = 174.24 X 10^3 mm/s^2. Since υ_o= 0, the phase is ϕ = $\tan^{-1}(\omega_n x_o/0)$ = π/2, or 90°. Hence, in this case, the response is

$$x(t) = 10 \sin\left(132t + \frac{\pi}{2}\right) = 10 \cos\ (132t) \text{ mm}$$

FORCED UNDAMPED VIBRATION



Let F be a sinusoidal force,

$$F = F_o sin\omega t$$

$$m\ddot{x} + kx = F_o sin\omega t$$

$$\ddot{x} + {\omega_n}^2 x = (F_o/m) sin\omega t$$

$$x = x_h + x_p$$

$$x_h = A sin\omega_n t + B cos\omega_n t$$

$$x_p = X_o sin\omega t$$

$$\dot{x}_p = X_o \omega cos\omega t;\ \ddot{x}_p = -X_o \omega^2 sin\omega t$$

$$-X_o \omega^2 sin\omega t + X_o {\omega_n}^2 sin\omega t = F_o sin\omega t$$

$$X_o = \frac{F_o}{{\omega_n}^2 - \omega^2}$$

when $\omega = \omega_n;\ X_o \to \infty$

This *condition is known as resonance and must be avoided.*

$$x = A sin\omega_n t + B cos\omega_n t + \frac{F_o}{{\omega_n}^2 - \omega^2} sin\omega t$$

Example 2

Compute and plot the response of a spring-mass system subjected to a force of magnitude 23 N, driving frequency of twice the natural frequency and initial conditions given by x_0 = 0 m and v_0= 0.2 m/s. The mass of the system is 10 kg and the spring stiffness is 1000 N/m.

Solution 2

First compute the various coefficients for the response

$$\omega_n = \sqrt{\frac{1000\ \mathrm{N/m}}{10\ \mathrm{kg}}} = 10\ \mathrm{rad/s},\ \omega = 2\omega_n = 20\mathrm{rad/s}$$

$$f_0 = \frac{23}{10} = 2.3\ \mathrm{N/kg}$$

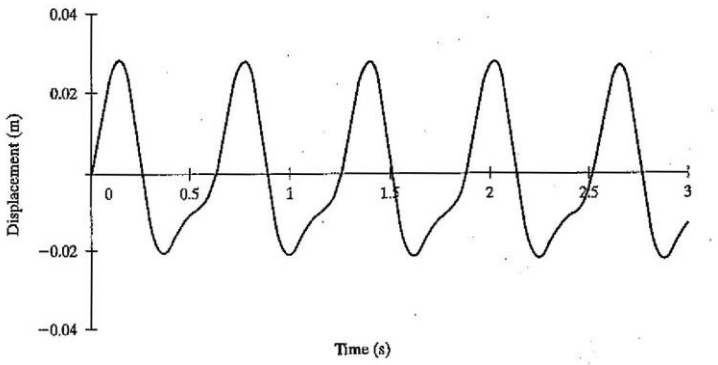
$$\frac{v_0}{\omega_n} = \frac{0.2\ \mathrm{m/s}}{10\ \mathrm{rad/s}} = 0.02\mathrm{m},$$

$$\frac{f_0}{\omega_n^2 - \omega^2} = \frac{2.3\ \mathrm{N/kg}}{(10^2 - 20^2)\mathrm{rad}^2/\mathrm{s}^2} = -7.9667 \times 10^{-3}\mathrm{m}$$

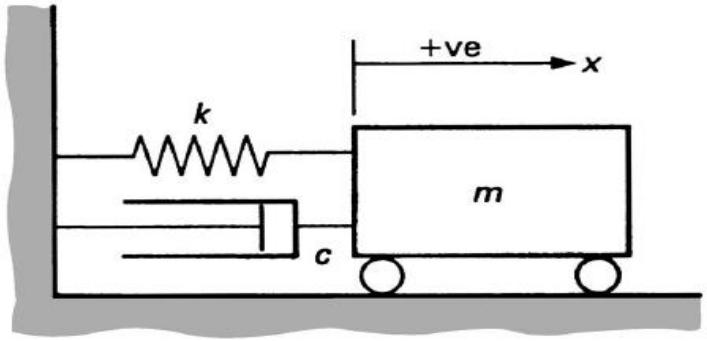
With these values, the response becomes

$$x(t) = 0.02\sin 10t + 7.667 \times 10^{-3}(\cos 10t - \cos 20t)\mathrm{m}$$

The plot of the time response is given in the Figure .



FREE DAMPED VIBRATION



$$m\ddot{x} + c\dot{x} + kx = 0$$

$$x = X_o e^{\lambda t}$$

$$\dot{x} = \lambda X_o e^{\lambda t}; \quad \ddot{x} = \lambda^2 X_o e^{\lambda t}$$

$$m\lambda^2 + c\lambda + k = 0$$

$$\lambda = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m}$$

$$d = c^2 - 4mk$$

$d = 0;$ *Critically damped system*
$d > 0;$ Overdamped system
$d < 0;$ Less damped − underdamped System

$$d = c^2 - 4mk = 0$$

Critical damping; $c_c = 2\sqrt{mk} = 2m\omega_n$

Damping ratio, $\xi = \frac{c}{c_c} = \frac{c}{2m\omega_n}$

$\Rightarrow \xi = 1;$ *Critically damped system*
$\xi > 1;$ Overdamped system
$\xi < 1;$ Less damped − underdamped System

$$\lambda = \frac{-2\xi m\omega_n \pm \sqrt{(2\xi m\omega_n)^2 - (2m\omega_n)^2}}{2m}$$

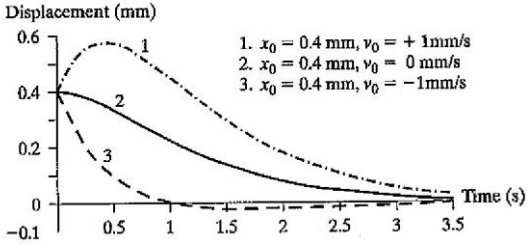
$$\lambda = -\xi\omega_n \pm \omega_n\sqrt{\xi^2 - 1}$$

FREE UNDAMPED VIBRATION

- $\xi = 1; Critically\ damped\ system$

$$\lambda_1 = \lambda_2 = -\omega_n$$

$$x(t) = (A + Bt)e^{-\omega_n t}$$

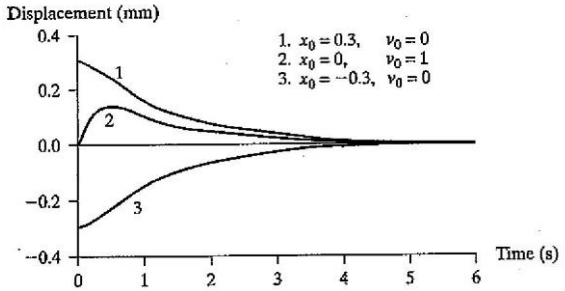


- $\xi > 1;$ Overdamped system

$$\lambda_1 = -\xi\omega_n - \omega_n\sqrt{\xi^2 - 1}$$

$$\lambda_2 = -\xi\omega_n + \omega_n\sqrt{\xi^2 - 1}$$

$$x(t) = e^{-\xi\omega_n t}(Ae^{\lambda_1 t} + Be^{\lambda_2 t})$$

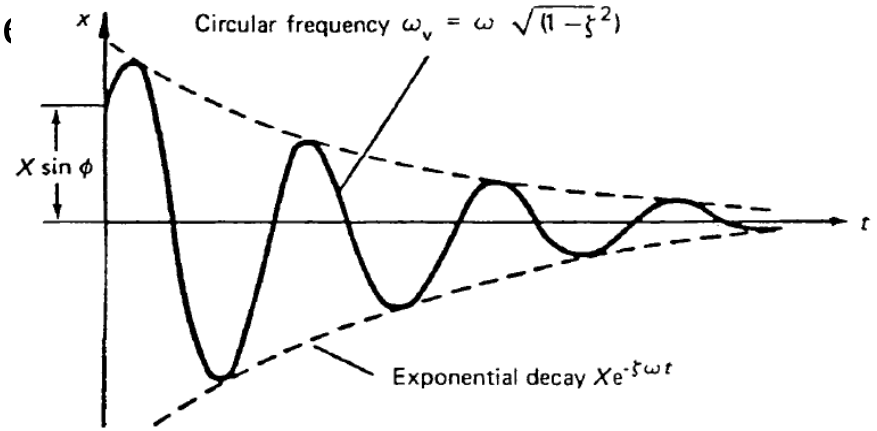


- $\xi < 1;$ Less damped − unde

$$\lambda_1 = -\xi\omega_n - \omega_n j\sqrt{1 - \xi^2}$$

$$\lambda_2 = -\xi\omega_n + \omega_n j\sqrt{1 - \xi^2}$$

$$x(t) = Ae^{-\zeta\omega_n t}\sin(\omega_v t + \phi)$$



Example 3

The damping rate of the spring is measured to be 0.11 kg/s. Calculate the damping ratio and determine if the free motion of the spring-bolt system is over-damped, under-damped, or critically damped. , *m* = 49.2 x 10−3 kg and *k* = 857.8 N/m

Solution.

Using the definition of the critical damping coefficient of equation

$$c_{cr} = 2\sqrt{km} = 2\sqrt{(2857.8\ \mathrm{N/m})(49.2 \times 10^{-3}\mathrm{kg}}$$

= 12.993 kg/s

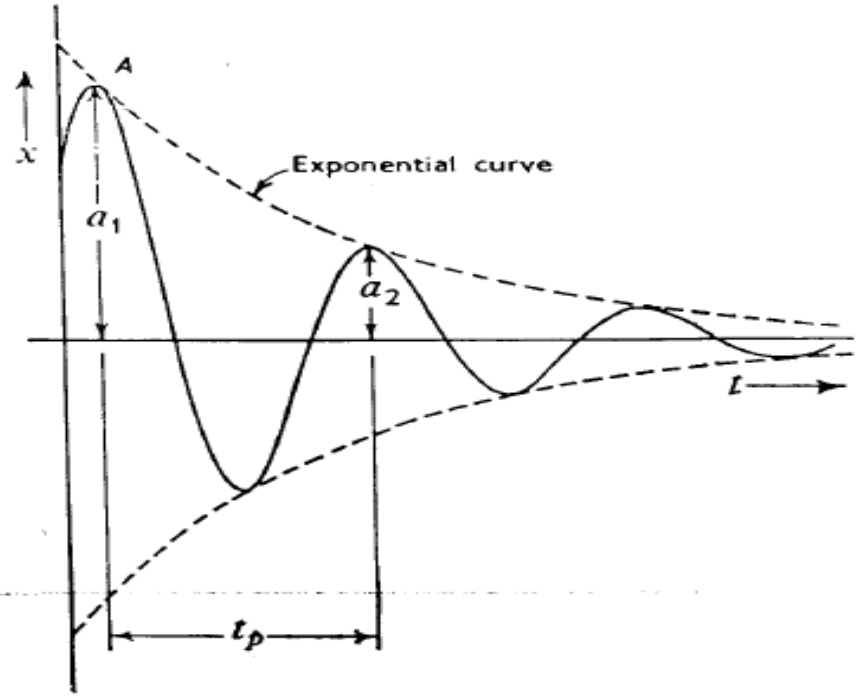
If *c* is measured to be 0.11 kg/s, the critical damping ratio becomes

$$\xi = \frac{c}{c_{cr}} = \frac{0.11\ (\mathrm{kg/s})}{12.993\ (\mathrm{kg/s})} = 0.0085$$

or 0.85% damping. Since ξ is less than 1, the system is under-damped. The motion resulting from giving the spring-bolt system a small displacement will be oscillatory.

Logarithmic decrement

- A convenient way of determining the damping in a system is to measure the rate of decay of oscillation.



$$a_1 = Ae^{-\xi\omega_n t}\sin(\omega_v t + \phi)$$

$$a_2 = Ae^{-\xi\omega_n t}\sin(2\omega_v t + \phi)$$

$$a_r = Ae^{-\xi\omega_n rt}\sin(r\omega_v t + \phi)$$

$$t = t_p = \frac{2\pi}{\omega_v}; r = 1, 2, 3 \ldots$$

$$\sin(r\omega_v t + \phi) = \sin(\omega_v t + \phi)$$

$$\frac{a_1}{a_r} = e^{\xi\omega_n t(r-1)}$$

$$ln\left(\frac{a_1}{a_r}\right) = \xi\omega_n t(r-1) = \frac{2\pi\xi(r-1)}{\sqrt{1-\xi^2}}$$

Example 4

A static deflection test is performed on a less damped system with a mass of 2 kg and the stiffness is determined to be 1.5 X 10^3 N/m. The displacements at t_1 and t_2 are measured to be 9 and 1 mm, respectively. Calculate the damping coefficient.

Solution

From the definition of the logarithmic decrement

$$\delta = \ln\left[\frac{x(t_1)}{x(t_2)}\right] = \ln\left[\frac{9\text{mm}}{1\text{mm}}\right] = 2.1972$$

From equation (1.65),

$$\xi = \frac{2.1972}{4\pi^2 + 2.1972^2} = 0.33 \text{ or } 33\%$$

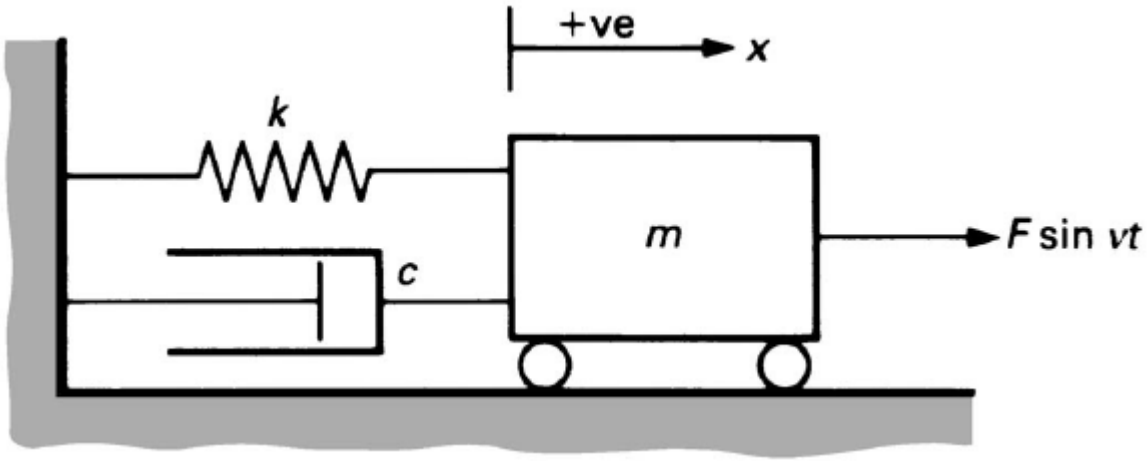
Also

$$c_{cr} = 2\sqrt{km} = 2\sqrt{(1.5 \times 10^3 \text{N/m})(2 \text{ kg})} = 1.095 \times 10^2 \text{ kg/s}$$

and from equation (1.23) the damping coefficient becomes

$$c = c_{cr}\xi = (1.095 \times 10^2)(0.33) = 36.15 \text{ kg/s}$$

FORCED DAMPED SYSTEMS



$$m\ddot{x} + c\dot{x} + kx = F_o sinvt$$

$$x = x_h + x_p$$

The homogenous solution, x_h depends on the value of ξ

The particular solution

$$x_p = X_o \sin(\omega t - \emptyset)$$

$$\dot{x}_p = X_o \omega cos(\omega t - \emptyset); \quad \ddot{x}_p = -X_o \omega^2 \sin(\omega t - \emptyset)$$

$$X_o = \frac{\mathrm{F_o}/m}{\sqrt{(\omega_\mathrm{n}^2 - \omega^2)^2 + (2\xi\omega_\mathrm{n}\omega)^2}} = \frac{\delta}{\sqrt{(1-\tau^2)^2 + (2\xi\tau)^2}}; \quad \tau = \omega/\omega_n$$

$$\emptyset = \tan^{-1}\left(\frac{2\xi\omega_\mathrm{n}\omega}{\omega_\mathrm{n}^2 - \omega^2}\right) = \tan^{-1}\frac{(2\xi\tau)}{(1-\tau^2)}$$

Example 5

Consider again the simple spring-mass system of consisting of a spring and a bolt. Calculate the value of the steady-state response if = 132 rad/s for f_0 = 10 N/kg. Calculate the change in amplitude if = 125 rad/s. , the natural frequency and damping ratio are given as 132 rad/s and 0.0085 respectively.

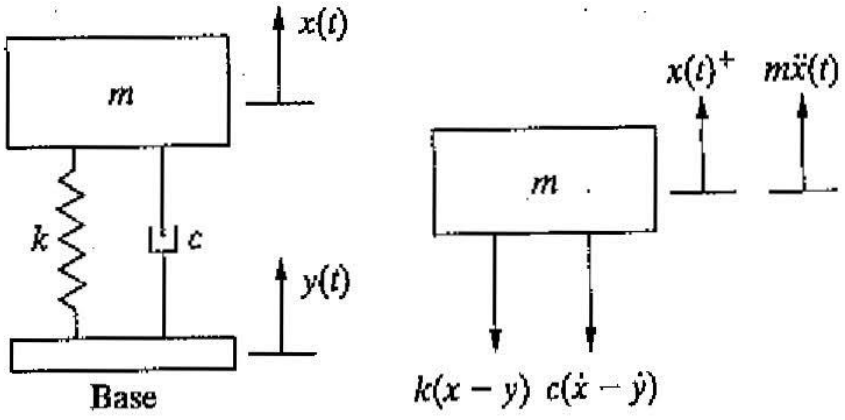
Solution

$$X_o = \frac{F_o/m}{\sqrt{(\omega_n^2 - \omega^2)^2 + (2\xi\omega_n\omega)^2}}$$

$$X_o = \frac{10}{\{[(132)^2-(132)^2]^2+[2(0.0085)(132)(132)]^2\}^{1/2}} = \frac{10}{2(0.0085)(132)^2}=0.034$$

If the driving frequency is changed to 125 rad/ s, the amplitude becomes 0.005 m. So a slight change in the driving frequency from near resonance at 132 rad/s to 125 rad/s (5%) causes an order-of-magnitude change in the amplitude of the steady-state response.

Base Excitation



EQUATION OF MOTION

$$m\ddot{x} + c(\dot{x} - \dot{y}) + k(x - y) = 0$$

$$y(t) = Y \sin\omega t$$

ABSOLUTE DISPLACEMENT, X

$$m\ddot{x} + c\dot{x} + kx = c\dot{y} + ky$$

$$\frac{X}{Y} = \sqrt{\frac{1 + (2\xi\tau)^2}{(1 - \tau^2)^2 + (2\xi\tau)^2}}$$

RELATIVE DISPLACEMENT, z

$$let\ z = x - y$$

$$m\ddot{z} + c\dot{z} + kz = -m\ddot{y}$$

$$\frac{Z}{Y} = \frac{\tau^2}{\sqrt{(1 - \tau^2)^2 + (2\xi\tau)^2}}$$

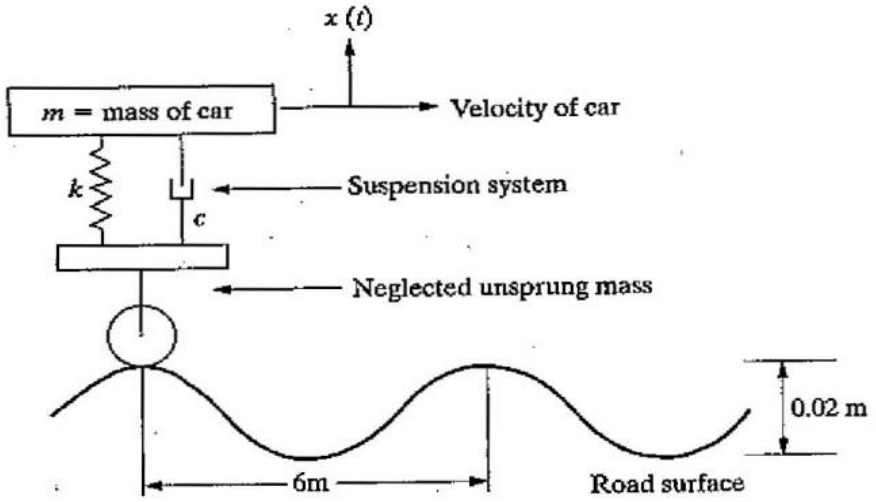
Example 6

A very common example of base motion is the single-degree-of-freedom model of an automobile driving over a rough road or an airplane taxiing over a rough runway, indicated below. The road is approximated as sinusoidal in cross section providing a base motion displacement of

$$y(t) = (0.01\ \text{m}) \sin \omega_\text{b} t$$

where $$\omega_\text{b} = v \left(\frac{\text{km}}{\text{h}}\right) \left(\frac{1}{0.006\ \text{km}}\right) \left(\frac{\text{hour}}{3600\ \text{s}}\right) \left(\frac{2\pi\ \text{rad}}{\text{cycle}}\right) = 0.2909\text{v rad/s}$$

and where *v* denotes the vehicle's velocity in km/h.



Solution

$$\omega_n = \sqrt{\frac{4 \times 10^5 \text{ N/m}}{1007 \text{ kg}}} = 19.93 \text{ rad/s}$$

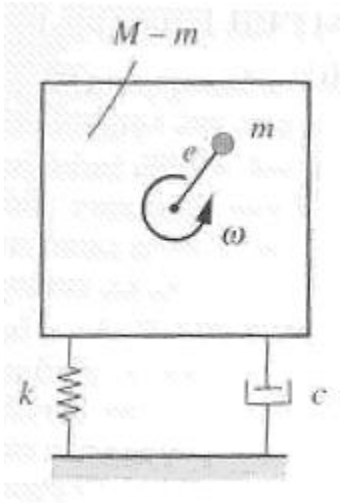
So that *r* = 5.818/19.93 = 0.292 and

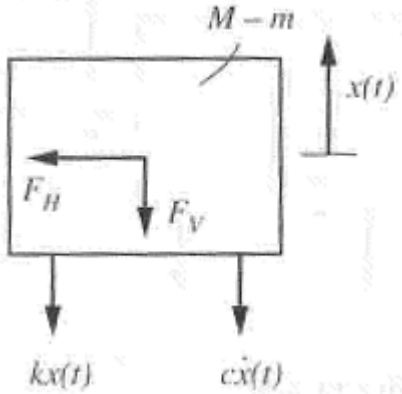
$$\xi = \frac{c}{2\sqrt{km}} = \frac{20{,}000 \text{ N. s/m}}{2(4 \times 10^5 \text{ N/m})(1007 \text{ kg})} = 0.498$$

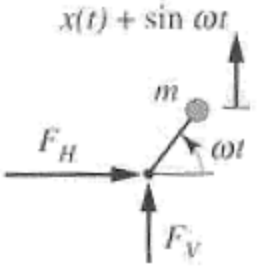
The deflection experienced by the car will be

$$X = (0.01 \text{ m}) \sqrt{\frac{1 + [2(0.498)(0.292)]^2}{[1 - (0.292)^2]^2 + [2(0.498)(0.292)]^2}} = 0.0108 \text{ m}$$

Rotating Unbalance







Displacement of Total Mass

$$(\mathrm{M}-\mathrm{m})\ddot{\mathrm{x}}+\mathrm{c}\dot{\mathrm{x}}+\mathrm{kx}=0$$

$$\mathrm{M}\ddot{\mathrm{x}}+\mathrm{c}\dot{\mathrm{x}}+\mathrm{kx}=\mathrm{m}\ddot{\mathrm{x}};\quad x=e\,\sin\omega\,\mathrm{t}$$

$$X=\frac{me}{M}\frac{\tau^2}{\sqrt{(1-\tau^2)^2+(2\xi\tau)^2}}$$

Force Transmitted to support

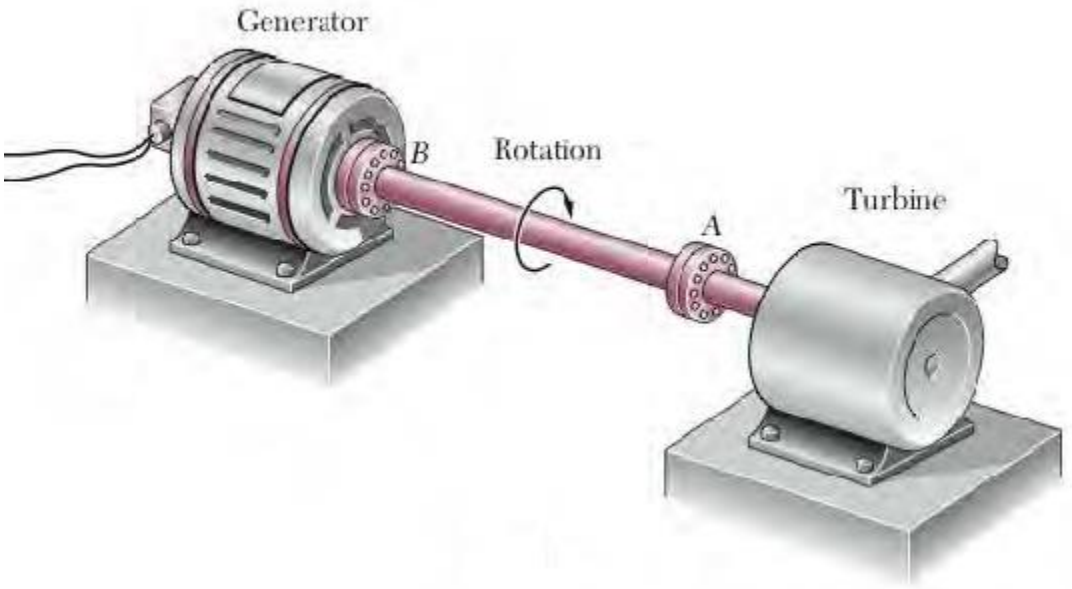
$$F_T=\mathrm{c}\dot{\mathrm{x}}+\mathrm{kx}$$

$$x=X_o\sin(\omega\,\mathrm{t}-\emptyset)$$

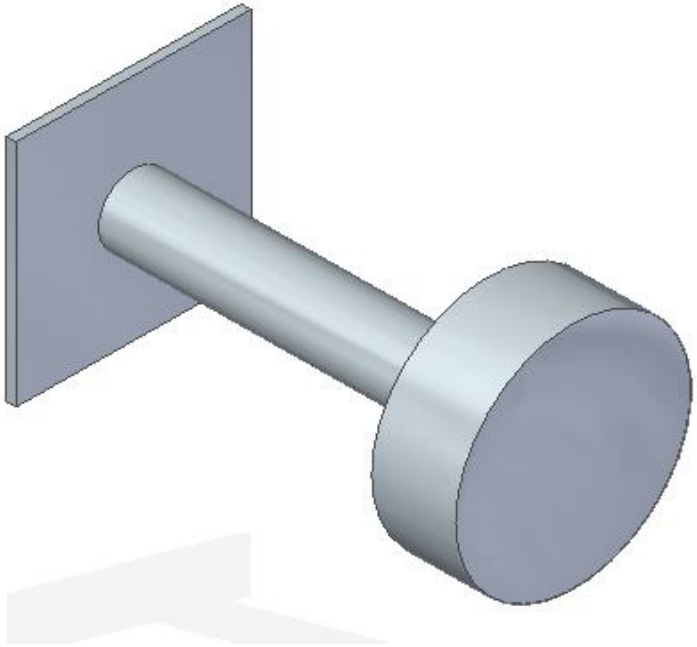
$$F_T=F_o\sqrt{\frac{1+(2\xi\tau)^2}{(1-\tau^2)^2+(2\xi\tau)^2}}$$

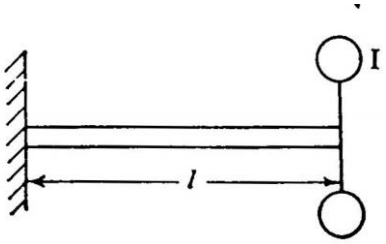
$$\emptyset=\tan^{-1}\frac{(2\xi\tau)}{(1-\tau^2)}$$

TORSIONAL SYSTEMS
SINGLE- AND MULTI- ROTOR SYSTEMS



Disc of moment of inertia, I, attached to shaft





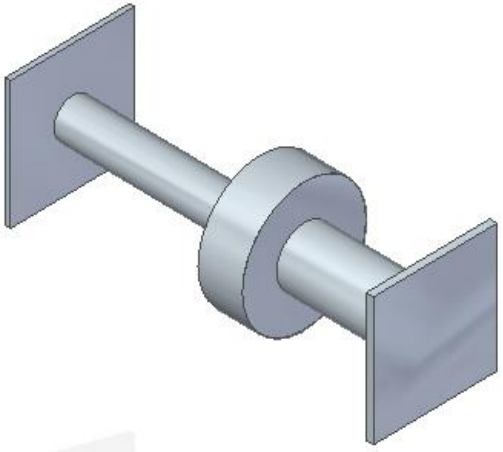
$$I\ddot{\theta} + k\theta = 0$$

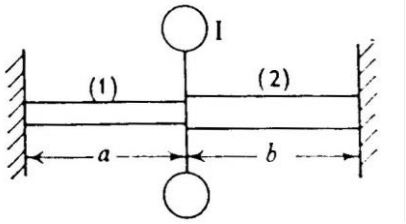
$$\ddot{\theta} + \omega^2\theta = 0$$

$$\omega^2 = \frac{k}{I}; \quad k = \frac{GJ}{l}; J = \frac{\pi d^4}{32}$$

$$\omega = \sqrt{\frac{GJ}{Il}}\,\text{rad/s}$$

Rotor attached to shaft between two fixed ends





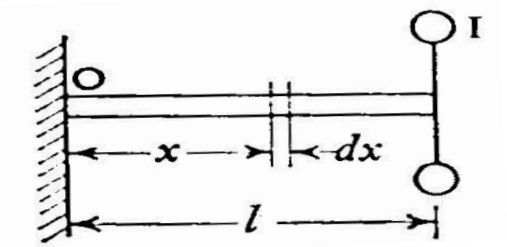
$$I\ddot{\theta} + (k_1 + k_2)\theta = 0$$

$$\ddot{\theta} + \omega^2\theta = 0$$

$$\omega^2 = \frac{k_1 + k_2}{I}; k_1 = \frac{GJ_1}{a}; k_2 = \frac{GJ_2}{b}; J_1 = \frac{\pi {d_1}^4}{32}; J_2 = \frac{\pi {d_2}^4}{32}$$

$$\omega = \sqrt{\frac{G}{I}\left(\frac{J_1}{a} + \frac{J_2}{b}\right)}\ \text{rad/s}$$

Effect of Inertia of Shaft; Torsionally Equivalent Shaft



Let the instantaneous angular velocity of the rotor in the above Figure be Ω.

Then angular velocity of element at distance *x* from O = $\frac{x}{l}\Omega$

If I_s is the polar moment of inertia of the shaft, then (kinetic energy) K.E. of element of length dx

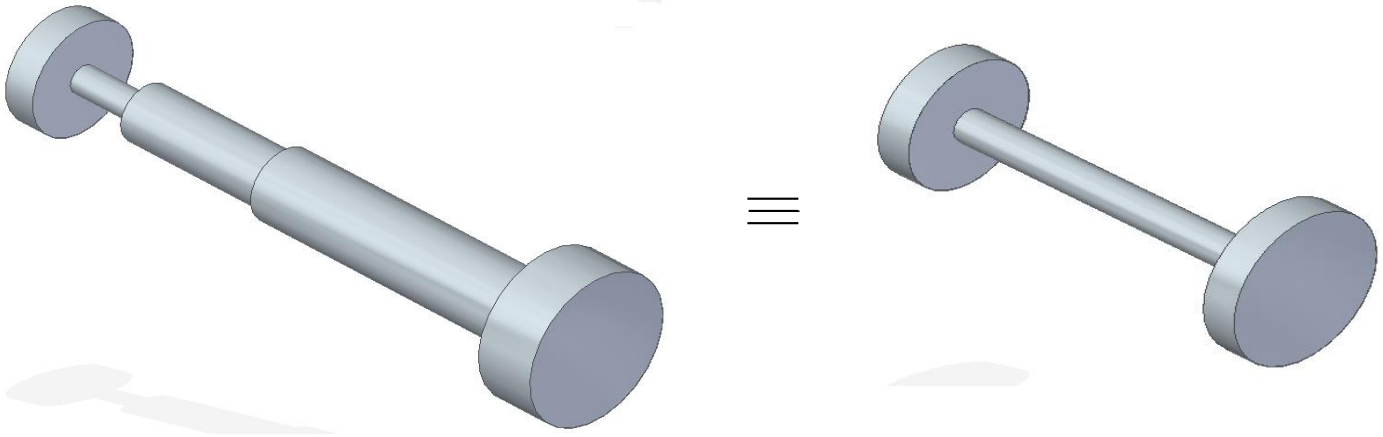
$$= \frac{1}{2} I_s \frac{dx}{l} \times \left(\frac{x}{l}\Omega\right)^2$$

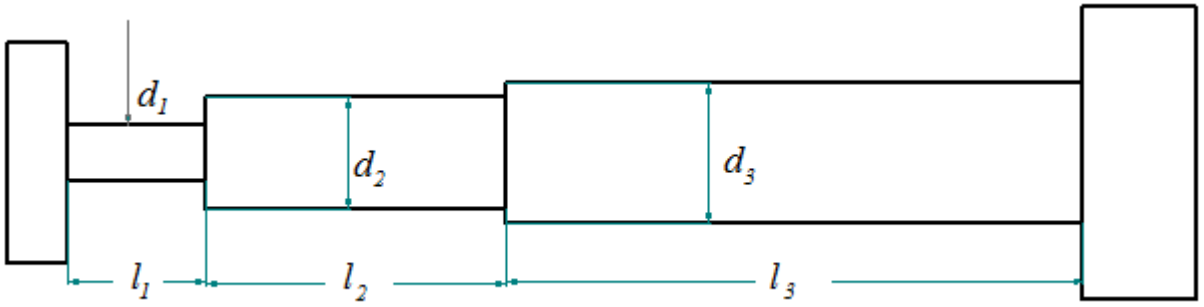
$$\therefore \text{ total K. E. of shaft} = \frac{I_s}{2l^3}\Omega^2 \int_0^1 x^2 dx = \frac{I_s}{2}\frac{\Omega^2}{3}$$

$$\therefore \text{ total K. E. of system} = \frac{1}{2}\left\{I + \frac{I_s}{3}\right\}\Omega^2$$

Shaft inertia may therefore be allowed for by adding a third of its inertia to that of the rotor.

Torsionally equivalent shaft
Step Shafts





$$\theta = \theta_A + \theta_B + \theta_C$$

$$\Rightarrow \frac{Tl}{GJ} = \frac{Tl_1}{GJ_1} + \frac{Tl_2}{GJ_2} + \frac{Tl_3}{GJ_3}$$

$$l = l_1\left(\frac{d}{d_1}\right)^4 + l_2\left(\frac{d}{d_2}\right)^4 + l_3\left(\frac{d}{d_3}\right)^4 = \sum_{i=1}^{n} l_i\left(\frac{d}{d_i}\right)^4$$

Example 1

A steel disc 300 mm diameter, of mass 29 kg, is suspended from the end of a wire 2·5 mm diameter and 1·5 m long which is clamped into a central hole in the disc, the upper end of the wire being rigidly supported. When the disc is set in torsional vibration it is found to make 10 complete oscillations in 78·2 s. Find the modulus of rigidity of the wire, and calculate the amplitude of the oscillation which may be allowed if the maximum permissible intensity of shearing stress in the wire is 140 MN/m^2.

Solution 1

Solution. For the wire,

$$J = \frac{\pi d^4}{32} = \frac{\pi \times 2.5^4}{32 \times 10^{12}} = 3.84 \times 10^{-12}\text{m}^4$$

For the disc,

$$I = \frac{mR^2}{2} = \frac{29 \times 0.15^2}{2} = 0.326\ \text{kgm}^2$$

$$n = \frac{1}{2\pi}\sqrt{\frac{GJ}{Il}}$$

i.e.

$$\frac{1 \times 10}{78.2} = \frac{1}{2\pi}\sqrt{\left[\frac{G \times 3.84 \times 10^{-12}}{0.326 \times 1.5}\right]}$$

Hence $G = 82.4\ \text{GN/m}^2$

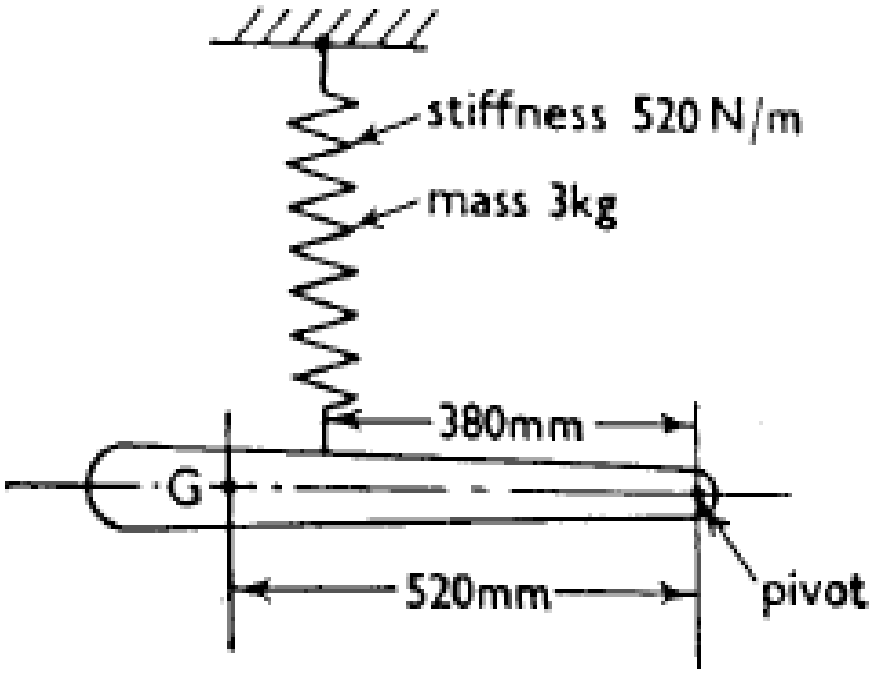
If τ is the shear stress in the wire at a radius *r*, then $\frac{\tau}{r} = \frac{G\theta}{l}$, i.e.,

$$\frac{140 \times 10^6}{0.00125} = 82.4 \times 10^9 \times \theta$$

$$\therefore \theta = 2.04\ \text{rad or } 117^0$$

ASSIGNMENT 1

The bar shown has a mass of 5 kg and the radius of gyration about the c.g. is 0.235 m. Find the natural period of oscillation.



ASSIGNMENT 2

A machine of mass 70 kg is mounted on springs and is fitted with a dashpot to damp out vibrations. There are three springs each of stiffness 9 kN/m and it is found that the amplitude of the vibrations diminishes from 36 mm to 6 mm in two complete vibrations. Assuming that the damping force varies directly as the velocity, determine the resistance of the dash-pot at unit velocity and compare the frequency of the damped vibrations with the frequency when the dash-pot is not in operation.

Assignment 3

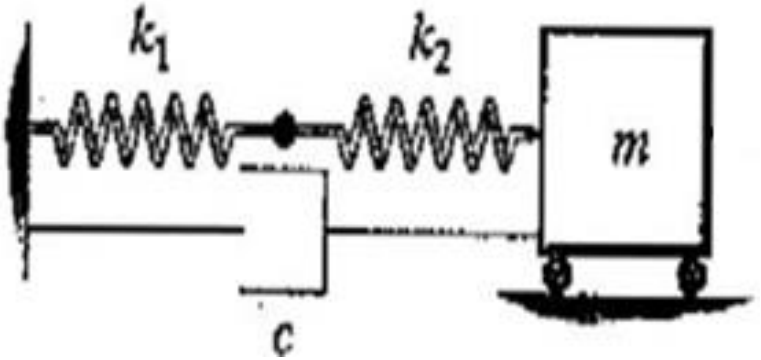
The figure below shows a model of a train coach.

i. Compute the Natural frequency of the system

ii. Calculate the Damped Natural frequency

iii. Classify the systems as over-damped , under-damped or critically damped and state the reason for your answer.

iv. Compute the response of the system below, given that xo=10 mm and vo=-20 mm/s. (for the values: k_1 = 2 k N/m, k_2 = 5 kN/m, c = 150 kg/s, and m = 50 kg)



Assignment

Obtain an expression for the natural frequency of torsional vibration of an elastic shaft carrying one wheel. For the system shown in the figure below, calculate the natural frequency. Specify the dynamic shear stress in the various portions of the shaft corresponding to an amplitude of vibration of 1□ at wheel A. The modulus of rigidity of the shaft material is 80GPa.

